

Assignment: NET Centric Computing

Course Code: CSC378

Topic: Unit 8 - Securing ASP.NET Core Application (Theory & Practical)

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Submission Instructions:

Please complete the following questions by combining theoretical explanations with practical implementations.

For practical components, include required commands, code snippets, configuration details, security implementation details, and screenshots of application execution and security testing results.

Questions:

1. Implementing Authentication Using ASP.NET Core Identity

Explain the concept of ASP.NET Core Identity and discuss its role in implementing authentication and user management in web applications.

Explain the process of adding authentication to an ASP.NET Core application and configuring Identity services.

Develop an ASP.NET Core MVC application to demonstrate user authentication using ASP.NET Core Identity.

Perform the following tasks:

1. Create an ASP.NET Core MVC application.
2. Add ASP.NET Core Identity packages and configure Identity services.
3. Configure Identity database, user registration, user login, logout functionality, and password management.
4. Implement user authentication features.
5. Configure authentication middleware.
6. Test valid and invalid authentication attempts.
7. Document Identity architecture, authentication process, configuration, and screenshots.

2. Implementing Authorization Using Roles, Claims, and Policies

Explain authorization in ASP.NET Core and discuss roles, claims, and policies.

Develop an ASP.NET Core application demonstrating role-based, claim-based, and policy-based authorization.

Tasks:

1. Create user roles such as Admin, Faculty, and Student.
2. Assign roles to users.
3. Implement role-based authorization.
4. Create user claims.
5. Configure authorization policies.
6. Secure controllers and action methods.

7. Demonstrate authorized and unauthorized access.
8. Document authorization implementation and screenshots.

3. Securing Controllers and Action Methods

Explain the importance of securing controllers and action methods.

Tasks:

1. Create multiple controllers.
2. Apply Authorize and AllowAnonymous attributes.
3. Secure specific action methods.
4. Implement access control based on authentication, roles, and policies.
5. Test controller access.
6. Document security mechanisms and output screenshots.

4. Demonstrating Common Web Application Vulnerabilities and Prevention

Explain common security vulnerabilities:

- Cross-Site Scripting (XSS)
- SQL Injection
- Cross-Site Request Forgery (CSRF)
- Open Redirect Attacks

A. Cross-Site Scripting (XSS) Attack

Create an input form, demonstrate malicious script injection, and implement prevention using input validation, output encoding, and sanitization.

B. SQL Injection Attack

Create a database login system, demonstrate SQL injection vulnerability, and prevent it using parameterized queries, Entity Framework Core, and input validation.

C. Cross-Site Request Forgery (CSRF)

Create a form submission system, demonstrate unauthorized requests, and implement anti-forgery token protection.

D. Open Redirect Attack

Create redirect functionality, demonstrate unsafe redirection, and implement URL validation and safe redirect methods.

Required Documentation:

- Introduction to ASP.NET Core Security
- ASP.NET Core Identity concepts
- Authentication implementation
- Authorization using roles, claims, and policies
- Securing controllers and action methods
- XSS attack and prevention
- SQL Injection attack and prevention

- CSRF protection
- Open Redirect prevention
- Source code
- Configuration details
- Screenshots of implementation and output results